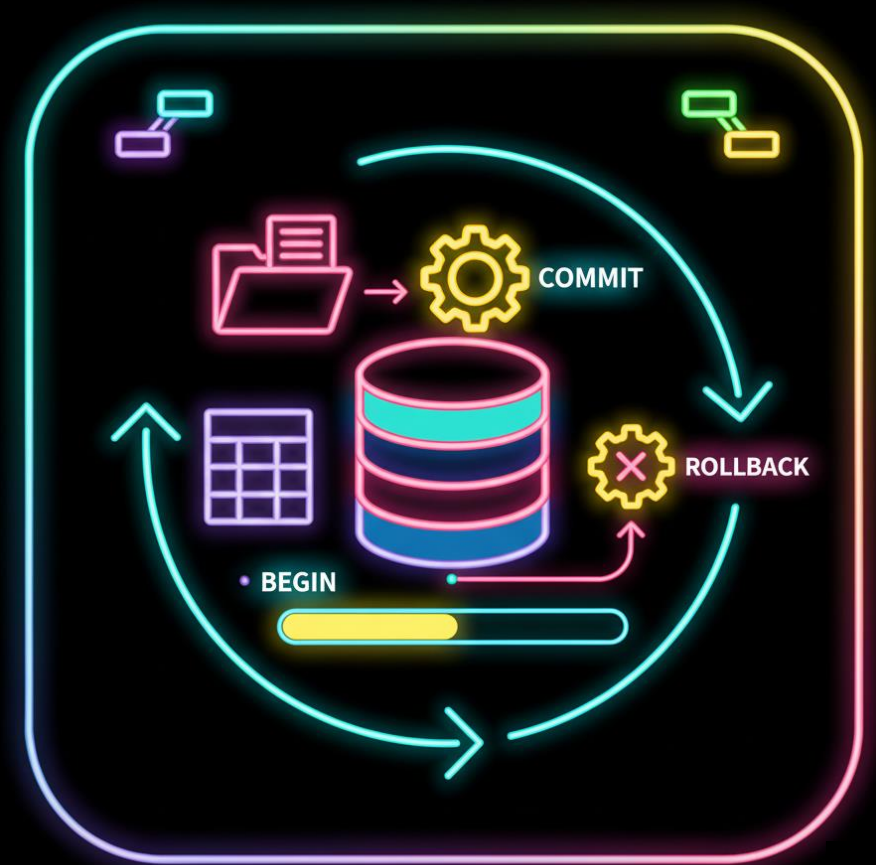


Programmatic SQL Objects

TCL - Understanding SQL Transactions and ACID Properties



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What is a Transaction & Why it Matters?

What is a Transaction?

- A transaction is a single logical unit of work in SQL
- Combines multiple statements into one operation
- Either all succeed or none succeed

Command	Purpose
BEGIN TRANSACTION	Start a transaction
COMMIT	Save changes permanently
ROLLBACK	Undo all changes since last BEGIN TRANSACTION
SAVEPOINT	Mark a point to partially rollback

Why Transactions Matter?

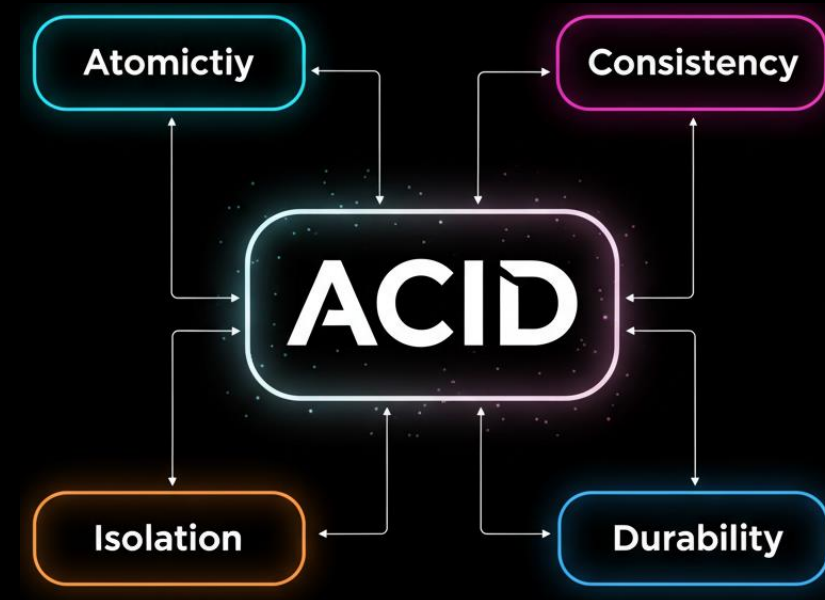
- Ensure data integrity
- Handle failures safely (power loss, crashes, errors)
- Allow multi-user databases to work consistently

ACID Properties

ACID Properties ensure your data stays:

- **Accurate**: Data remains correct and free from corruption
- **Reliable**: Transactions are processed consistently and predictably
- **Safe**: Data integrity is maintained, even in case of failures

Designed for complex, multi-user environments



Property	Meaning	Why It Matters
Atomicity	All or nothing. Either all statements execute, or none do.	Prevents partial updates.
Consistency	Transaction moves the database from one valid state to another.	Keeps data accurate and valid.
Isolation	Transactions run independently without interfering with each other.	Avoids conflicts in multi-user systems.
Durability	Once committed, changes persist, even after a crash.	Data is safely stored.